

# FreeRTOS - Assignment 05

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## 1. LED Manager Gatekeeper

- Three tasks request LED operations.
  - Task A -> LED ON
  - Task B -> LED OFF
  - Task C -> LED TOGGLE
- Only one task should control GPIO.
- Requirements
  - Create LED Gatekeeper Task.
  - Tasks send LED commands through queue.
  - Gatekeeper Task performs GPIO operations.

## 2. Smart Room Monitoring System

- Design a multitasking application using:
  - LDR Sensor
  - LM35 Temperature Sensor
  - UART
  - LED
- Requirements
  - Task 1: Light Monitoring Task
    - Read LDR value every 500 ms.
      - If room becomes dark:
        - LED ON
      - Otherwise:
        - LED OFF
  - Task 2: Temperature Monitoring Task
    - Read LM35 every 1 second.
    - Calculate temperature.
    - If temperature exceeds 40°C:
      - Warning LED Blink
  - Task 3: UART Reporting Task
    - Every 2 seconds print:
      - Temperature = xx C
      - Light = xxx
      - LED State = ON/OFF